AI 101 for Mechanical Engineers · Module 16

# A small project you can defend

## Lesson 16.2

# What you will learn

> Frame a bounded task; document inputs, target, provenance and operating domain; keep training separate from locked evaluation data; and report baselines, errors and physical checks reproducibly.

# Lesson overview

Module 16 · Lesson 2

## A small project you can defend

AI 101 for Mechanical Engineers

### 1 Bound the task and document the data

**Example:** predict measured electrical input power (kW) from the flow rate and pressure rise available at decision time.

- Decision and intended user
- Target and inputs available when the decision is made
- Operating domain: bounds, parameters, load ranges
- Data source, units, measurement context, assumptions

Synthetic data must be labelled. It can exercise a workflow but cannot validate a real machine.

### 2 Keep training and evaluation apart




Split sizes are illustrative.

- Fit preprocessing on training data only
- Use validation data for development choices
- Use the locked test set once, for the final evaluation

Records from the same machine, event, design or time sequence stay together. Random row splits can look too optimistic.

### 3 Compare and report with units

- Baseline 1: training-mean predictor
- Baseline 2: physical calculation, where appropriate

Hydraulic power = pressure rise × flow rate; electrical input power also needs efficiency and losses.

**Regression:** MAE and RMSE in kW; signed residuals, e.g. errors growing at high pressure.
**Classification:** class counts, precision, recall.

Add a known case or independent physical check. Metrics do not prove validity outside the evaluated conditions.

### 4 Keep evidence for each stage, and keep the work your own

| | |
|---|---|
| **Framing** | decision, inputs, provenance |
| **Development** | split, baseline, fitted preprocessing |
| **Evaluation** | locked-test errors, limitations |
| **Verification** | known-case and independent physical checks |

Start from a starter notebook, explain changes in your own words, keep it runnable from a clean kernel, and disclose AI help as policy requires. Do not send restricted data to external services.

**Key takeaway**

**One bounded decision, a locked test, fair baselines and physical checks.**
A reviewer should be able to tell a repeatable method from an unsupported conclusion.

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# Bound the project to one decision

- Options: pump power, bearing scores, operating profiles, calculation critique
- Example: predict measured electrical input power in kW
- Inputs: flow and pressure rise available at decision time
- State the decision, user, target, inputs and useful evidence

# Hydraulic power is not electrical input power

## Hydraulic power

- Pressure rise × volumetric flow

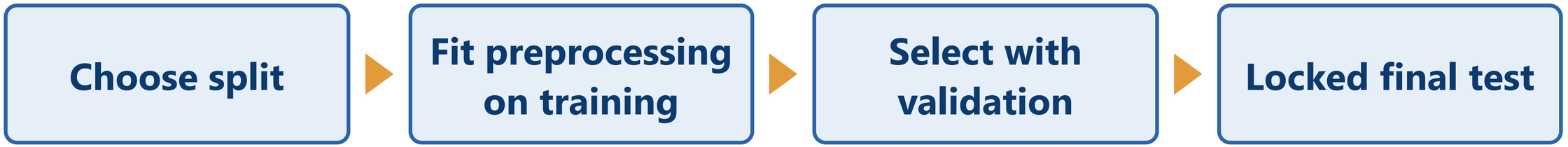
## Electrical input power

- Also requires accounting for efficiency and losses

# Document the operating domain and data

- Define bounds, operating parameters and load ranges
- Record source, units, measurement context and assumptions
- Use approved measurements when available
- Label synthetic data clearly; it cannot validate a real machine

# Keep evaluation data out of development




- Choose a split that reflects intended use [54]
- Keep related machine, event, design or time records together
- Random row splitting can make evaluation too optimistic

# Compare baselines on the same cases

### Training-mean predictor

- A simple reference

### Physical calculation

- A second comparison, when appropriate

A baseline helps show whether the model adds value beyond a simple or established approach.

# Report errors with units on the locked test

| Task | Report |
| --- | --- |
| Regression | MAE and RMSE in the target’s unit |
| Regression residuals | Signed errors in consistent SI units |
| Classification | Class counts; precision and recall |

> Residuals can reveal systematic under- or over-prediction, e.g. errors growing at high pressure.

# Metrics need physical interpretation and checks

- Interpret results in engineering terms, not as a score alone
- Add an independent physical check or known case
- Metrics do not establish validity outside evaluated conditions
- Retain evidence for framing, development, evaluation and verification

# Keep the notebook reproducible and your own

- Starters: data inspection, pump power, fault detection, draft verification
- Inspect assumptions; explain changes in your own words
- Run from a clean kernel; record data, environment, split, choices [36]
- Disclose AI use; send no private or restricted data externally without authorization

> A complete reference is one route, not a substitute for your own work.

CHECK YOUR UNDERSTANDING

# Write one project sentence containing a decision, inputs, an output and evidence.

Answer before moving on.

## CHECK YOUR UNDERSTANDING

**Write one project sentence containing a decision, inputs, an output and evidence.**

> Help judge pump efficiency: predict measured electrical input power (kW) from flow and pressure rise at decision time; compare locked-test MAE and RMSE with a training-mean baseline and physical calculation.

CHECK YOUR UNDERSTANDING

# Why can a random row split make results look too optimistic?

Answer before moving on.

CHECK YOUR UNDERSTANDING

## Why can a random row split make results look too optimistic?

> Related records from the same machine, event or time sequence can end up in both training and test sets, so the test no longer reflects new conditions.

# Key takeaways

- Bound the project to one decision; document inputs, target, provenance, domain
- Label synthetic data; it cannot validate a real machine
- Fit preprocessing on training data only; keep the test locked
- Compare with training-mean baseline and, where appropriate, physical calculation
- Report errors with units, add a physical check, stay reproducible

# References

**[36]** J. Pineau et al., “Improving reproducibility in machine learning research (A report from the NeurIPS 2019 reproducibility program),” J. Mach. Learn. Res., vol. 22, no. 164, pp. 1–20, 2021.

**[54]** D. R. Roberts et al., “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” Ecography, vol. 40, no. 8, pp. 913–929, 2017. doi: 10.1111/ecog.02881

Numbers match the course reference list. The written lesson is the authoritative text.